



Aror University of Art, Architecture, Design & Heritage Sukkur.

Department of Artificial Intelligence and Multimedia Gaming

Programming for AI (Spring-2026)

LAB No. 02

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Objective of Lab No. 02:

After performing lab 2, students will be able to:

- Understand and Apply conditional statements (if, if-else, if-elif)
- Understand and Apply Loops (for, while, nested Loops)
- Use Break and Continue Statement

Lab Tasks:

Task 1: Simple If Statement

Scenario: Check if a person is eligible to access a library membership.

- Ask the user for their age.
- Print "Eligible for library membership" if age ≥ 12 .
- Print nothing otherwise.

Task 2: If-Else Statement

Scenario: Check if a person is eligible for a gym membership.

- Take the user's monthly attendance as input.



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- Print "Membership Active" if attendance ≥ 10 , else print "Membership Inactive".

Task 3: If-Elif Statement (Single Choice)

Scenario: Temperature advisory.

- Input temperature in Celsius.
- Print advisory:
 - 35°C : "Heatwave Alert!"
 - $25\text{--}35^{\circ}\text{C}$: "Warm Weather"
 - $15\text{--}24^{\circ}\text{C}$: "Pleasant Weather"
 - $<15^{\circ}\text{C}$: "Cold Weather"

Task 4: If-Elif Statement (Multiple Choices)

Scenario: Movie ticket pricing.

- Input age of a person.
- Print ticket price:
 - Age < 5 : Free
 - Age 5–12: \$5
 - Age 13–59: \$12
 - Age ≥ 60 : \$8

Task 5: For Loop

Scenario: Multiples Finder.

- Input a number N.



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- Print all multiples of 3 from 1 to N.

Task 6: For Loop with Conditional

Scenario: Odd-Even Number Classifier.

- Input a number N.
- Print all numbers from 1 to N along with "Odd" or "Even".

Task 7: While Loop

Scenario: Guess the Number Game.

- Set a fixed number (e.g., 7).
- Keep asking the user to guess the number until they get it right.
- Print "Correct!" when guessed correctly.

Task 8: Loop with Break

Scenario: Exit on keyword.

- Keep asking the user to enter words.
- Stop the loop if the user types "exit".
- Print all words entered before "exit".

Task 9: Loop with Continue

Scenario: Skip invalid inputs.

- Input 10 numbers from the user.
- Ignore negative numbers using continue.



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- Print the sum of all valid numbers.

Task 10: Nested Loop Application Scenario: Attendance Record Checker.

- Input the number of students and the number of days.
- For each student, input their attendance (P for present, A for absent) for each day.
- At the end, print each student's total days present and absent.